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INVITED COMMENTARY

# Acute:Chronic Workload Ratio: Conceptual Issues and Fundamental Pitfalls

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The number of studies examining associations between training load and injury has increased exponentially. As a result, many new measures of exposure and training-load-based prognostic factors have been created. The acute:chronic workload ratio (ACWR) is the most popular. However, when recommending the manipulation of a prognostic factor in order to alter the likelihood of an event, one assumes a causal effect. This introduces a series of additional conceptual and methodological considerations that are problematic and should be considered. Because no studies have even tried to estimate causal effects properly, manipulating ACWR in practical settings in order to change injury rates remains a conjecture and an overinterpretation of the available data. Furthermore, there are known issues with the use of ratio data and unrecognized assumptions that negatively affect the ACWR metric for use as a causal prognostic factor. ACWR use in practical settings can lead to inappropriate recommendations, because its causal relation to injury has not been established, it is an inaccurate metric (failing to normalize the numerator by the denominator even when uncoupled), it has a lack of background rationale to support its causal role, it is an ambiguous metric, and it is not consistently and unidirectionally related to injury risk. ***Conclusion:*** There is no evidence supporting the use of ACWR in training-load-management systems or for training recommendations aimed at reducing injury risk. The statistical properties of the ratio make the ACWR an inaccurate metric and complicate its interpretation for practical applications. In addition, it adds noise and creates statistical artifacts.

***Keywords:*** injuries, training load, training principles, critical analysis

The number of studies examining associations between training load (TL) and injury has increased exponentially in recent years. One of the most popular TL metrics in the current literature, embraced by practitioners and researchers, is the acute:chronic workload ratio (ACWR).[1] ACWR has been studied in many sports and has also been promoted in an International Olympic Committee consensus statement.[2] It is now being implemented widely to manipulate TL even though there is, in our opinion, a lack of theoretical, observational, or experimental evidence for its use in this regard. The ACWR metric appears to be the tip of the iceberg with regard to the one-size-fits-all metrics that are now commonly being applied to numerous sports and any kind of injury (eg, contact, noncontact, injuries to various body parts, etc). Although the declared nature of these studies is always descriptive and sometimes predictive, the associations found between ACWR and injury risk have often been used to suggest or recommend that TL be manipulated in order to reduce the risk of injuries. However, in order to suggest that ACWR can be modulated to effect positive change, it is necessary to demonstrate via counterfactual causal analyses that modifying ACWR alters injury or performance outcomes.[3–6] Such a counterfactual causal analysis approach could provide support for a cause–effect relation that is based on well-established statistical regression methodologies—a method that has never been employed in ACWR research.[7] Furthermore, the absence of a specific conceptual framework or hypothesis about the mechanisms linking generalized metrics (eg, ACWR) to injuries may generate investigations with counterintuitive and inconsistent results that are difficult to reconcile with reasonable injury mechanisms. Given the absence of studies supporting its causal effect, manipulation of ACWR should not be recommended. Before disseminating and implementing a system broadly, stronger evidence, coming from methodologically but also conceptually sound studies, is needed. In this commentary, we detail additional conceptual and methodological issues making the ACWR an unlikely candidate causal prognostic factor and also showing that it is an inaccurate metric for its practical implementation.

## Lack of Conceptual Basis

Pure prognostic factors are often identified through biological reasoning (candidate approach) but can also be identified through data-driven methods that are unrelated to hypothesized causal mechanisms.[8] Unlike pure prognostic research, however, exploring the causal effect of a potential factor on an outcome (eg, whether altering TL induces a change in injury risk) requires a causal structure for understanding the appropriate statistical analysis and interpretation.[9–11] This structure is commonly developed based on expert knowledge. Since its initial presentation, the ACWR[1] has been grounded on the Banister (impulse-response/fatigue-fitness) model[12] as a framework explaining the causal role of ACWR for injury: "Conversely, if acute workload exceeds the chronic workload (ie, the athlete has performed inadequate training to develop 'fitness' or workloads have been rapidly increased resulting in 'fatigue'), then the athlete is considered underprepared and likely at an increased risk of injury (see Figure 1). As such, the acute:chronic workload ratio indicates both the athlete's risk of injury and preparedness to perform." The acute load, calculated as the average

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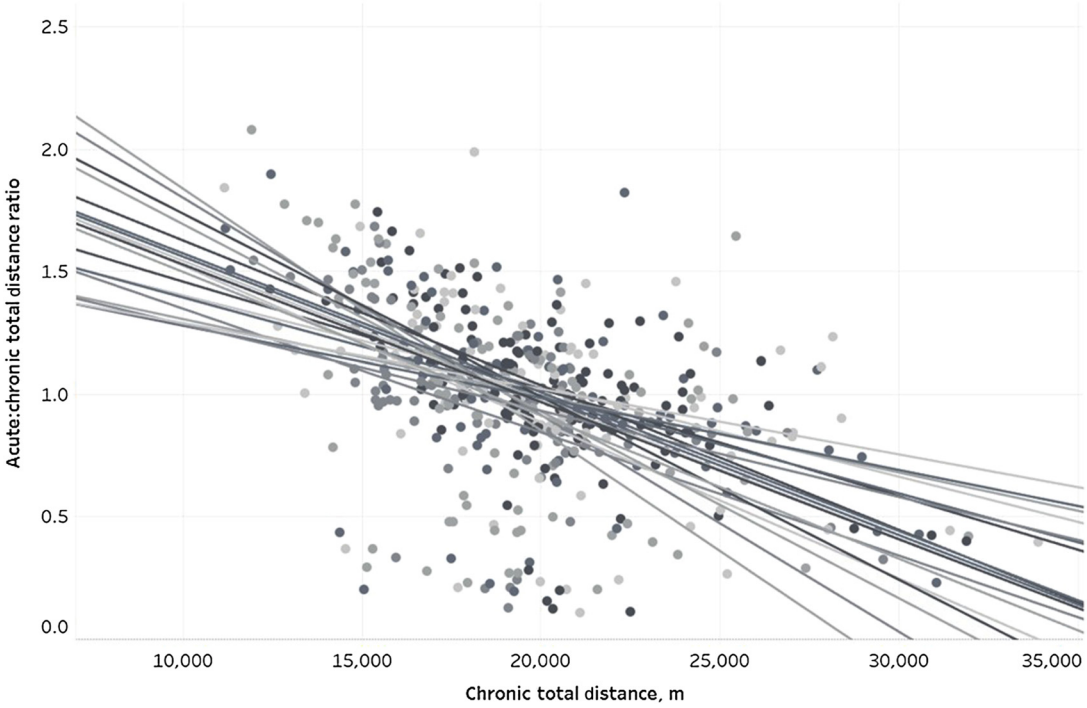


**Figure 1** — Within-subject correlations between acute:chronic workload ratio and chronic load in 20 Australian Football League players with more than 24 data points each (for 1 season) calculated using total distance, as in the study by Lolli et al[21] (–0.52, 95% CI, –0.10 to –0.78). CI indicates confidence interval.

of the TL accumulated in the last week (commonly the week preceding an injury), has been suggested to reflect the fatigue component (negative training effect) of the Banister model, while the average of the last 4 weeks of TL, defined as the chronic load, should represent fitness (positive training effect). The ACWR has also been related to the training stress balance calculated from the difference between the fatigue and fitness components, again based on the Banister's model.[1] The time windows for the calculation of these variables are based on an approximation of the time decays that are used in the original model to weight the sum of the TLs (training impulse).[12] These averaged time windows clearly do not reflect the time decays of the original model concept mathematically or conceptually. In addition, the time decays are calculated by modeling each individual and are not set a priori and applied at a group level (same values for all). Furthermore, there are no reasons or explanations given for selecting 1 week (for acute) or 4 weeks (for chronic) (or whatever time frame) as an approximation of the time decays. The extension of Banister's model from performance to injury occurrence should be based on a biological rationale coupling the mechanisms explaining performance to the mechanisms explaining injuries. The corollary to this coupling in ACWR assumes that an improvement in performance would correspond an increase in injury risk, and vice versa. This is clearly unreasonable, and even Banister's original model has been criticized for the lack of a physiological underpinning, which makes it difficult to link this model to mechanisms of injuries.[13] Recent adaptations have been proposed in an attempt to bring the ACWR back to the time decays proposed in the original Banister model (eg, an exponentially weighted moving average for calculating the acute and chronic loads).[14] However, although this adaptation is conceptually stronger and more consistent with the dissipation time of the negative and positive components of the original model, it does not compensate for the lack of physiological, biological, or biomechanical rationale behind the acute and chronic load time windows. Furthermore, the addition of the exponentially weighted moving average to ACWR still does not make it mathematically the same as Banister's original model. Given the lack of any justifications, previous studies have tried several windows, commonly focusing on, discussing and attributing a meaning to the time windows showing some significant associations, that is, post hoc justification. However, these justifications are usually practical; that is, the authors decided the lengths of the windows based on scheduling (eg, based on microcycles of 1 wk: 6 training days plus 1 d of match).[15] Because the adaptation time of the different tissues is independent from competition and training scheduling, these justifications do not look reasonable (eg, different tissues have different time course adaptations/maladaptations) and do not provide any information for developing causal structure or any other etiological consideration. The time windows used are usually completely subjective and lack transparency in the selection process. If these windows are manually modified post hoc until they work with the data, there are 2 possible scenarios, as follows: (1) in the best case scenario, the study is exploratory in nature; however, no causal explanation or structure has been proposed after 5 y and approximately 130 papers on the topic; and (2) the worst case scenario is that it represents (hopefully unintentional) *P* value hacking and/or model overfitting. Furthermore, the ACWR is applied to a myriad of internal and external TL measures (perceived exertion, distance, balls bowled, accelerometer-based measures, etc) that assume that all these measures are related to similar mechanisms or have a similar effect on injuries (type, location, and nature, such as contact and noncontact). This unreasonable assumption is, for example, shown by the U-shaped ACWR model proposed in editorials and consensus statements that combine data


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from 3 different sports and 3 different measures of TL.[2,16–18] In addition, studies have shown that ACWR in the same population has different relations with injuries depending on the selected TL measures.[19] Finally, the original Banister model (or the training stress balance) is additive and *not* a ratio. An underlying rationale or explanation has never been provided why a ratio model should be used instead of an additive model. Unfortunately, the unfounded use of a ratio presents a fundamental problem in itself, as we discuss below.

## Empirical Considerations on the Ratio and Its Fundamental Pitfalls

Among the first criticisms related to the ACWR were the mathematical coupling and the use of a ratio.[20,21] Use of simulated data from a reference study revealed the presence of mathematical coupling between acute load and chronic load, because the more recent week of load is typically included within the 4-week period calculation.[20] The problem of mathematical coupling inherent in the conventional ACWR calculation is, nevertheless, only one fundamental aspect that should be considered by researchers and practitioners when interpreting and using the ratio as a causal prognostic factor for manipulating TL. This problem can be easily fixed by uncoupling the calculation. However, once the uncoupling is performed, a formal appraisal of whether a simple ratio provides meaningful relative measures of TL for prognostic purposes is still required.[22] In the presence of a substantial association, simple ratios are well-recognized for failing to normalize the numerator variable (acute load) consistently for the denominator variable (chronic load).[23] In a follow-up study, the independence between the uncoupled terms of the ACWR showed that ratio normalization was unnecessary and, paradoxically, introduced nonphysiological noise in the acute load data, thereby compromising a reliable quantification of TL in relative terms.[21] This was not peculiar to the data used by Lolli et al,[21] but it reflects a well-known problem of ratios, including the risk of creating artifacts.[24]

To further confirm this, in the present commentary, we have provided the same analysis presented by Lolli et al[21] but using data from a different sport and different metrics. We used the external load data relative to total distance (TD), high-speed running (HSR), and sprinting distance (SPR) from 41 professional Australian Football League players over 43 weeks (preseason and in-season data). After uncoupling the chronic and acute loads to eliminate mathematical coupling,[21] we calculated the associations between ACWR and chronic load using all data collected. We only excluded the ACWR corresponding to x/0 and 0/0. We found large negative individual associations for all the external load measures (Table 1). However, because Lolli et al[21] used only ACWR calculated with 4 complete measurements (weeks) for the chronic load (ie, no 0 values), we deleted the data not fitting this criterion. This resulted in the loss of 40% of the data, which further highlighted concerns about the feasibility of this metric in practical settings.[25] Nevertheless, in order to compare our results with those of Lolli et al[21] (see Figure 1), we used the remaining data to calculate the relation between ACWR and chronic load for players with more (n = 20, high group) or less (n = 21, low group) than 24 data points. Again, we found moderate to large correlations consistent with the one shown by Lolli et al[21] (–0.50; 95% confidence interval [CI], –0.18 to –0.71). These results show that regardless of the preanalysis treatment of the data, the association between ACWR and chronic load is omnipresent.

In agreement with the previous analysis,[21] the results presented in this commentary confirm that the ACWR, even when uncoupled, fails to normalize the numerator by the denominator. These results highlight the inappropriateness of using a ratio (in addition to the concerns related to the acute and chronic calculation). This was expected because the problems created by the ratios have been known for more than a century[26] and addressed in seminal papers and textbooks published in the 1940s[27,28] and, more recently, by other authors.[23,24,29,30] While some recent papers have focused on the issue of coupling,[31–33] they did not recognize that this is just a secondary problem to the more critical issue of using a ratio. Ratio and coupling are both problematic; resolving the problem of the coupling still leaves the problem of the ratio, and investigation is needed in order to ascertain its appropriateness. Indeed, as highlighted by Lolli et al,[21] there are 2 conditions that must be met in order for a ratio to provide meaningful information, as follows: (1) there must be a *true* meaningful association between numerator and denominator, and (2) the ratio must normalize the numerator by the denominator.[24] The first editorial[20] dealt with the coupling problem, because any relation between numerator and denominator should be *true* and not spurious (as for the coupled). In the second editorial[21] as well as in the current commentary, ACWR data were uncoupled in order to meet condition (1). Having done this, the ACWR failed to normalize the acute load by the chronic load, underlining the inappropriateness of using this ratio (ie, it failed condition [2]). As a consequence, examining correlation between coupled and uncoupled, as has been done in some studies,[31–34] has nothing to do with the issues raised by Lolli et al[20,21] and reinforced by us.

As indirect confirmation of the inaccurate normalization, previous authors have purposely deleted chronic loads below 1 or 2 SDs (or *z* scores) from their statistical analyses to avoid spikes in the ACWR.[15,35,36] The decision to remove these datapoints is questionable but mainly surprising. Indeed, a ratio is used when the researchers believe that the important factor is the numerator, but they want to control it for differences in the variable at the denominator.[24] But the necessity to further control by cutting some values of the chronic load means that the normalization did not work. In other words, they used a ratio to normalize the numerator by the denominator; this ratio introduced a problem, because it

**Table 1 Correlations Between Acute:Chronic Workload Ratio and Chronic Load in 41 Australian Rules Football Players**

| | | Total distance | | High-speed running | | Sprinting | |
|---|---|---|---|---|---|---|---|
| | n | *r* | 95% confidence interval | *r* | 95% confidence interval | *r* | 95% confidence interval |
| All | 41 | –.76 | –.51 to –.87 | –.52 | –.25 to –.71 | –.60 | –.36 to –.77 |
| High (>24 data points) | 21 | –.52 | –.10 to –.78 | –.40 | –.05 to –.72 | –.46 | –.20 to –.64 |
| Low (<24 data points) | 20 | –.50 | –.09 to –.77 | –.47 | –.05 to –.75 | –.49 | –.00 to –.73 |



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failed to normalize the numerator by the denominator; then they adopted an arbitrary solution (excluding data) in order to solve the problem they created by generating the ratio. Additional adjustments have been made by using the chronic load as an effect-measure modification, which is used to examine the effects of a variable for different levels of another variable, which is supposed to modify its effect.[19,36] However, this other variable (chronic load) is the one already supposedly adjusted by the ratio being the denominator of the ratio. Therefore, this stratification of the effects of the numerator for a variable used to adjust the numerator is a contradiction and, hence, makes no sense. In addition, chronic load cannot be an effect-measure modification only, but it is at least also a confounder, because it affects the ACWR value, again being the denominator of the ratio. The reality is that the ratio and, specifically, the ACWR fail to normalize the acute by the chronic.

There are several conceptual problems generated by the ratio, and the consequent adjustments used in some literature are quite evident. In addition, given the difficulty in interpreting the roles of its components (which are what we are supposed to manipulate), all of these problems call into question the validity of ACWR as a potential causal prognostic factor other than as a reference metric (in practical setting). The concerns regarding the ratios are well-known and shared among the scientific community.[23,24,27–30]

## Inconsistent Results

One defense of ACWR is that, despite the conceptual and statistical problems, the literature has shown consistent relations with injury risk. First, even if this were true, without estimating causal effects, the manipulation of TL using ACWR or any other metric would be inappropriate and not supported by the findings and nature of the studies. However, without addressing the methodological aspects of these studies, we also contend that the relationship between ACWR and the risk of injuries is inconsistent, and the associations often appear in different directions. If, as an example, we examine studies published this year (2019) in this journal, Ahmun et al[37] showed, between ACWR and probability of injuries, a negative association (session RPE in cricket)—that is, the more you increase the acute load, the lower the injury risk. Johnston et al[38] showed a positive association (session RPE in endurance sports), indicating that the more you increase the acute load, the higher the injury risk. Raya-Gonzalez et al[39] showed no association (odds ratio; session RPE in football), indicating that whatever the change in the acute load, there was no associated increase or decrease in injury risk. Hulin et al[40] reported a likely higher (> 75% using the magnitude-based inference) injury rate at ACWR > 1.9 only (PlayerLoad in a rugby league). Finally, a study by Lathlean et al[41] very recently showed an inverted U-shape between ACWR and injury (ie, the more you increase or decrease the acute load, the lower the injury risk). This study and many others contradict the popular and accepted U-shaped ACWR model suggesting that exercising outside a range of about 0.8 to 1.3 (the so-called "sweet spot") increases injury risk.[17,18] This commonly accepted U-shaped relation has been recently questioned (with a request of retraction),[42] given methodological mistakes such as errors in the calculations of the risks, combination of injury rates taken from measures of exposures calculated from different TL indicators (session RPE, balls bowled, and running total distance), combination of different sports, use of unpublished data, arbitrary assignment of rates calculated for categories to single ACWR values and treated as continuous, assignments of rates to values outside the ranges for which they were calculated, no formal statistical analysis, and selective reporting/modification of previous data.[42]

It is clear that the association between ACWR and injury is inconsistent, and the reasons for the heterogeneity are difficult to explain. According to GRADE (Grading of Recommendations, Assessment, Development and Evaluations) adaptation for prognostic studies,[43] the level of evidence supporting its use even as a prognostic factor should be downgraded. Furthermore, the inconsistent and unexpected (counterintuitive) associations may also suggest that the findings are due to chance, misclassification of the predictor, selection bias, mixing of effects (confounding), intervention effects, and heterogeneity.[44] Lack of sound scientific rationale combined with unconvincing replicability and inconsistent epidemiologic effects may also indicate that the associations are not causal but coincidental.[45] Despite these concerns, national athlete management systems, educational courses, and software now utilize this metric as a basis for recommendations on how to adjust TL in order to reduce injuries. We contend that these recommendations are based on descriptive and not even predictive studies (no valid predictive model is available in the literature). Before implementing something, the quality of the evidence and the validity of the measures should be examined carefully and critically. Unfortunately, once such guidelines have been implemented on a large scale, dismissing the tool is more challenging, given the cognitive dissonance it may generate and consequent protective reactions. This can favor a switch to a dogmatic approach. Scientists should challenge dogmas, not fuel or create them.

## Ambiguous Metric

In addition, we believe that this ratio is deceptive, because it moves the attention away from the real meaning of the numbers provided. Those applying this model did not recognize that the U-shaped model suggests that after a tapering or recovery week, injury risk increases, which is clearly not reasonable. However, despite all of the issues we have noted, the sweet spot illustrated in opinion articles (editorials) has been published in 2 consensus articles (one in the present journal[16] and one by the Olympic Committee,[2] where it became a "validated model"). Its application also ignores that the ACWR is based on retrospective analyses with studies using an injury lag. In other words, the majority of the studies have examined whether the TL completed in the 4 to 5 weeks before an injury provides an estimation of injury risk. However, the training completed between the last week of training and injury occurrence (typically 1–7 d, but studies have used a longer lag) is not used and considered in the analysis, which assumes that this factor has no effect. From a practical point of view, this means that once you know your athlete is at higher risk, whatever training you carry out after the week included in your analysis, the injury risk remains high (sometimes called delayed effect). Therefore, once you have recognized this risky situation, there is nothing you can do in terms of TL manipulation. Again, even assuming a causal effect, from a practical point of view, it would not be very useful. The only way to use ACWR for preventing injury would be to plan training so that athletes remain at an acceptable ACWR level. But the acceptable level is not clear, because an ACWR for total distance (as an example) of 2.0 (meaning doubling the acute TL) has been associated in professional players with higher injury risk (eg, Bowen et al[19]) but associated with no change in increase injury risk and even a trend toward lower risk (Jasper et al,[46] ie, another example of inconsistency). From the available literature using the ACWR, it is impossible to determine what is a safe increase when planning progression for total distance in professional soccer players.


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Furthermore, experienced fitness coaches, athletes, and sport scientists rarely plan for large spikes in TL. Few experienced coaches would purposely double, triple, or quadruple TL from 1 week to another. Following periods of reduced TL (e.g. recovery weeks) the ACWR is automatically increased. This computational issue may be solved erroneously by also reducing TL in the weeks after the planned recovery, and this may lead to undertraining. Therefore, the analysis and identification of these spikes are mainly retrospective; most of the time, they are likely due to uncontrollable contextual factors. However, as mentioned previously, there is no way to manipulate TL to address this risky situation, even assuming a cause-effect relation that has been never actually shown. These practical problems have never been discussed or considered despite the use of AWCR for reducing injury risk being commonly advocated in studies and even in international guidelines and recommendations.

## Conclusion

The ACWR and its components (acute and chronic) present several critical issues that suggest that it is unlikely to be an appropriate candidate for a causal prognostic factor and is an inaccurate metric to be implemented in practice. There is not a sound background rationale linking this metric to causal injury mechanisms; it is an inaccurate metric (fails to normalize), it moves attention away from the real meaning of the numbers on which is based, and it is not consistently related to injury risk. We focused on the ratio problem, because using a ratio that fails to normalize can create statistical artifacts, including the obfuscation of any potential pathways between load and injury occurrence or vice versa.[24] While this may be not so relevant for predictive purposes, it is important for etiological research and for using or suggesting ACWR for reducing injury risk. Any ratio index included as an exposure variable in a model precludes a clear understanding of the actual contribution of the numerator or denominator variables on clinical outcome. In this context, ratio normalization of acute for chronic load may introduce unwanted noise and have an unappreciated potential to confound the true level of an exposure variable (ie, acute load) from an etiological standpoint.

A very recent study (submitted preprint: https://osf.io/preprints/sportrxiv/e8kt4/) has shown that the ACWR creates statistical artifacts, misclassification, and miscalibration.[47] The chronic load is not reflective of the transient level of preparation of an athlete but is just a rescaling procedure, magnifying the effect estimates of the actual explanatory variable (ie, acute load). In addition, the use of the chronic load, which is simply a smoothing of the acute load values, reduces the coefficient of variation of the numerator (acute load) inflating the association with injury risk. A critical appraisal of ACWR has been also published during the review process of this commentary, confirming our concerns, and underlining additional critical issues.[48] The conceptual and methodological arguments (supported by computational and statistical analyses) against the use of ACWR have been, therefore, presented from independent research groups. There is now a clear case for recommending that this flawed metric be disregarded in TL injury research.

## Practical Applications

- Proposing to manipulate the ACWR (and TL, accordingly) for reducing risk of injury is not supported from a conceptual, methodological, and practical point of view. As such, ACWR should also not be included in training load management systems or training recommendations aimed at reducing injury risk.
- Given the needs of daily applied practice, use of absolute measures of TL might be more informative for practitioners regarding the potential utility of different measures of TL for injury risk reduction (if any). In other words, we should focus on data with a real meaning and not, as it happened for the ACWR, building evidence with potentially flawed data.[49]
- The only practical alternative is to return, in the absence of a proven better alternative, to applying the usual training principles (eg, overload progression) and using TL (including any athlete feedback) to monitor whether the planned program, load, and progression have been completed by the athletes. To date, there is no quantitative way to know whether manipulating TL and its progression can increase or decrease the injury risk.

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